

Intrinsic-Dimension-Guided Bottleneck Selection Revisited: A Common-Protocol Comparison with Existing Criteria

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Abstract

Selecting a variational autoencoder's latent width requires distinguishing representation diagnostics from reconstruction requirements. We compare intrinsic-dimension guidance, ascending and grid searches, FONDUE variants, and local pruning adaptations on four image datasets with five candidate training seeds. A validation anchor defines an explicit reconstruction target; final ordinary VAEs provide a common dimension-transfer comparison. We report test error, target attainment, termination outcomes and recorded training costs, with missing overhead identified. At the primary MNIST setting, ID guidance and ascending search select the same six-dimensional model, with training accounts of 715 and 182 seconds. Shared reference-bank checks reject ID guidance on Fashion-MNIST, dSprites and CIFAR-10, which therefore use a coarse-search fallback. Selected dimensions vary with regularisation context, but method orderings reverse across datasets. The ARD adaptation did not vary its selection-stage weight and cannot support a zero-sensitivity claim. Active-unit counts depend on their threshold, and a retrospective paired random-feature comparison does not establish an AU-specific predictive benefit. These results delimit what ID and activity diagnostics alone establish about capacity under the tested architectures, quality targets and implementation choices.

Keywords: variational autoencoder; latent dimension; intrinsic dimension; active units; model selection; reconstruction quality; reproducibility

1. Introduction

The latent width of a variational autoencoder (VAE) [1] controls the capacity available to its representation. Choosing that width by validation over a candidate grid is straightforward, but requires training several models. Alternatives include intrinsic-dimension (ID) estimation [2,3], comparisons between sampled and mean latent representations [4], activity diagnostics [5,6], and pruning through reconstruction constraints or relevance priors [7,8]. These signals answer different questions: geometric dimension, coordinate use, reconstruction quality and network width need not coincide.

This study asks whether ID guidance reduces the recorded training cost of finding a small VAE that meets a stated reconstruction target, whether AU diagnostics provide information beyond other recorded quantities, and how selection changes with regularisation and readout settings. We compare selection rules using the same candidate-model families, data splits and quality yardstick. For pruning approaches, we explicitly evaluate *dimension transfer*: a returned count is mapped to a grid width and used to select an ordi-

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nary VAE. This comparison does not measure the performance of the original pruned or hierarchical-prior model.

The contribution is a conditional empirical comparison and an audit of what its measurements support. Four image datasets and five candidate-training seeds are represented. The regularisation analysis uses five weights, but its intervention differs for the pruning adaptations, as specified below. We provide numerical quality targets, test results, learning curves, paired readout analyses, and the executed ID-guided procedure. We distinguish algorithmic termination, fallback, quality attainment and compression, and separate measured training time from unmeasured overhead.

At the primary MNIST setting, ID guidance and ascending search return the same six-dimensional model, with recorded training accounts of 715 and 182 s, respectively. This supports a narrow conclusion about that implementation and target. The other results describe dependence on representations, training settings and diagnostic thresholds; they do not establish a universal ordering of selection methods. Experimental details are given in Sections 3–5, results in Section 6, and limitations in Section 7.

2. Related Work

2.1. Dimension, Reconstruction and Latent Activity

PCA [9], Isomap [10] and locally linear embedding [11] construct low-dimensional representations. ID estimators instead quantify local dimension from sample geometry. We use TwoNN and an MLE cross-check; other approaches include DANCo, expected simplex skewness and geodesic entropic graphs [12–14]. Estimator limitations and dependence on the sampled distribution motivate cross-checks [15,16]. Graph-based methods are outside this experiment’s scope; we do not infer that they would resolve or reproduce the observed disagreements. Lower ID in learned representations has also been observed in deep networks [17,18].

FONDUE [4] searches for the capacity preceding a separation between ID estimates of sampled and posterior-mean representations. Its appendix gives FONDUE-VAR, which chooses a count from latent-variable types. AU and variable types share an interest in unused coordinates but are different statistics: AU thresholds the across-data variance of posterior means, whereas the type classifier uses moments of posterior variances [6]. Agreement of their final counts is not equivalence of their stability or information content.

GECO with L_0 regularisation optimises a gated VAE under a reconstruction constraint, with KL and gate-count terms [7,19]. The source uses ARM; our adaptation uses hard-concrete gates [20]. Neither the nonconvex optimisation nor our implementation guarantees a global minimum capacity. Our validation anchor plus tolerance is a particular way of setting an external quality yardstick, not an identical implementation of the source’s constraint.

ARD-VAE uses a hierarchical relevance prior [8]. The source tunes a regularisation weight, uses a Gaussian approximation for its closed-form KL, and selects axes explaining 99% of a Jacobian-weighted variance score. It is therefore not threshold-free. The local adaptation has additional differences, including a finite-step decoder score and a fixed selection-stage weight; these are detailed in Appendix B.1.

2.2. Relation to the Conference Paper and Earlier Manuscript

Our NOLTA2025 contribution [21] concerned the relation between input ID and latent-space capacity. The available conference manuscript describes class-specific MNIST experiments with four fully connected and two convolutional architectures and four seeds. CNNs and multiple seeds are consequently not new features of this revision. The conference manuscript available for comparison has a working title different from the published title;

the reference uses the title and pages listed by the authors’ institutional publication record. Table 1 distinguishes this earlier scope from the present comparison. MAKE47 denotes the earlier journal submission, not a separate publication.

Table 1. Scope of the conference study, earlier journal submission and present revision.

Item	NOLTA2025	MAKE47	Present revision
Question	Input and latent ID versus capacity	ID-guided choice with AU verification	Selection quality, cost and diagnostic limitations
Models	Four FC and two CNN structures; MNIST classes; four seeds	FC/CNN and synthetic/image extensions	Fixed candidate families on four image datasets; five candidate seeds
Comparators	Architecture and capacity comparisons	Capacity sweeps and diagnostic checks	FONDUE variants, pruning adaptations, grid criteria and ascending search
Readouts	ID estimates	AU-based operational checks	Same-checkpoint type/AU trajectories and paired predictive analysis
Evidence	Conference settings	Earlier loss convention	Corrected-loss records; explicit targets, final test metrics and accounting limits

3. Evaluation Framework

3.1. Models and Quality Target

We distinguish the *ID-reference AE*, which supplies a representation for estimation, the *quality anchor*, an ordinary VAE at the largest candidate width, and the *selected model*, the ordinary VAE evaluated after dimension selection. Write $\mathcal{M}$ for the candidate grid, s for a training seed, and $D_s(m, \beta)$ for validation per-pixel MSE of the deterministic reconstruction $g(\mu(x))$. The implemented target is

$$m_a = \max \mathcal{M}, \quad T_{D,s}(\beta) = D_s(m_a, \beta) + \max\{0.10D_s(m_a, \beta), 0.001 \operatorname{Var}(X_{\text{val}})\}. \tag{1}$$

The variance is computed over all validation-array entries. The relative term dominates in every recorded image condition, giving $T_{D,s} = 1.10D_s(m_a, \beta)$. The configuration originally also specified a three-seed-SD term, but the executed target function was supplied one anchor at a time and omitted that term. We report the implemented rule; we do not retrospectively describe it as a pooled-seed target. Appendix A lists every dataset- β anchor and target.

For a fixed seed and β , the target is held constant during search. It is recomputed from a different anchor when β changes. A β -sweep therefore changes both the learned representation and the quality yardstick. It is not a fixed-quality intervention. A method that returns a valid width need not meet this external target; quality attainment is evaluated separately for every final model.

3.2. Cost Accounting and Shared Candidates

Candidates with the same dataset, architecture, likelihood, width, seed, epoch budget, stopping option and β are shared through a cache. Each selection rule is charged the recorded training duration of every distinct candidate it requests. Reusing a candidate within the rule costs no additional training time. Consequently, identical selected candidates have identical metrics across methods. They are paired observations, not inde-

pendently retrained method-specific models. A selected candidate already trained during search is reused; a short-search candidate receives the ordinary final training schedule.

The original ledgers did not charge the quality anchor to every method. For this revision, its recorded duration is added wherever absent; if it is also the final ordinary VAE, it is counted once. The resulting C_{train} includes candidate training, ID-reference AE training, the anchor, and separately recorded probe time where used for selection. Training timers include epoch-wise validation, but omit several post-training evaluations and ID computations. The pruning timers have slightly different boundaries, including their final readout. Thus C_{train} is an attributed recorded-time account, not an end-to-end wall-clock measurement. Exact overhead cannot be recovered from these logs. FONDUE epoch-calibration time is reported separately because it is performed once per dataset- β condition. No complete wall-clock speedup is claimed.

3.3. Outputs and Termination

Each trial records a raw returned dimension, the width of the evaluated model, the ID reliability decision where applicable, fallback use, the logged termination state, the budget flag, quality attainment, and whether the final width is smaller than the anchor. These events can overlap. In particular, fallback can produce a compact model that meets the target. The label SUCCESS in an older log indicates ordinary termination; it is not itself evidence of target attainment.

A budget-exceeded trial contributes its consumed time and, if it has a returned model, that model's dimension and metrics. A raw dimension below one is reported as degenerate; its time and failure count remain, while MSE is missing because no final model exists. The original implementation checks budgets after completed calls and can still obtain a final model. The reported limits are therefore soft accounting limits, not hard interruptions. Outcome tables preserve these distinctions.

3.4. Executed ID-Guided Procedure

Algorithm 1 describes the executed selection rather than a hypothetical improved variant. The reference bank uses three widths and three fixed reference seeds, shared across the five candidate seeds. Let $\bar{d}_r$ be the mean TwoNN estimate at reference width r , $\hat{d} = \text{mean}_r \bar{d}_r$, and $\bar{d}_{\text{MLE}}$ the mean MLE estimate over that same bank. The checks are

$$R = \frac{\max_r \bar{d}_r - \min_r \bar{d}_r}{\text{median}_r \bar{d}_r} \leq 0.15, \quad A = \frac{|\hat{d} - \bar{d}_{\text{MLE}}|}{\hat{d}} \leq 0.25. \quad (2)$$

The implemented initial window includes every grid point at or below the first grid point greater than or equal to $2\hat{d}$, not only points between $\hat{d}$ and $2\hat{d}$.

Algorithm 1 (Executed ID-guided selection and external evaluation).

Input: splits, $\mathcal{M}$, seed s , β , 10% target margin, floor coefficient 0.001, input TwoNN d_0 , reference seeds $\{42, 123, 777\}$, thresholds $(0.15, 0.25)$, and budgets in Table 2.

1. Obtain the quality anchor and fix $T_{D,s}$ from Equation (1). In a fallback run this is an external audit step, not a constraint used by the coarse selector; its training is charged.
2. For $k \in \{4, 8, 16\}$ set $r_k = \text{round}(kd_0)$ (nearest integer; ties to even, no grid snapping). Train an FC-AE at each width for each reference seed, and compute TwoNN and MLE on the fixed 10,000 training examples. Compute R , A and $\hat{d}$.
3. If either check fails, flag ID-unreliable and evaluate the coarse subset. Its zero-based indices are the unique values of $\text{round}(\text{geomspace}(1, |\mathcal{M}|, 4)) - 1$. Return its minimum-validation-MSE candidate and continue to step 6.
4. Otherwise set $m_0 = \min\{m \in \mathcal{M} : m \geq 2\hat{d}\}$, or m_a if the set is empty. Initialise the evaluated set with the anchor. Evaluate all remaining $m \leq m_0$ in ascending order, stopping after a completed call that exceeds the budget.
5. Return the smallest target-satisfying width in the evaluated set. If it is the anchor, record no compact alternative. The anchor is always acceptable; the code's planned expansion conditional on no acceptable evaluated candidate is consequently never reached.
6. Obtain or reuse the ordinary VAE at the returned width. For $m < 1$ return a degenerate output with no model. Retain budget flags even when a model is returned. Record AU as a diagnostic without changing m .
7. Return raw and final widths, model metrics, ID diagnostics, fallback and budget flags, quality/compression indicators and the time account.

The unreachable expansion branch limits the implemented ID-guided selector. It is disclosed rather than silently repaired while retaining old outcomes. The MNIST primary selection occurs within the initial window, so the reported comparison there does not require expansion. The earlier V1/V2 activity checks are not selection branches in this revision. Ascending search uses the same anchor and target, scans upwards to the first acceptable candidate, and requires no ID-reference AEs.

4. Theoretical Background

For an ideal smooth d -manifold, Whitney's embedding theorem gives an embedding into $\mathbb{R}^{2d}$ [22]. Exact continuous reconstruction through an injective encoder entails an embedding, but neither a noisy finite-sample ID estimate nor an approximate neural reconstruction satisfies these assumptions automatically. The multiplier two is therefore a heuristic under evaluation, not a quality guarantee or a lower bound inferred from measured ID.

In linear Gaussian latent-variable models, informative directions depend on the covariance spectrum and noise scale; analyses of linear VAEs and posterior collapse make this relation explicit [9,23]. VAE alignment with principal directions has also been studied [24]. These results motivate measuring how regularisation affects coordinate use. They do not equate AU with intrinsic or minimum embedding dimension in a nonlinear VAE. AU also depends on the latent coordinates and the threshold used to count them.

5. Experimental Setup

5.1. Data, Architecture and Optimisation

Table 2 gives the actual image-data settings. MNIST [25], Fashion-MNIST [26] and CIFAR-10 [27] use random subsets of the official training and test pools. A permutation with seed 20260912 selects 20,000 training-pool examples; another with seed 20260913 divides them 90:10. The 3000 test examples use seed 20260914. For dSprites [28], which has no official train/test split, the first 20,000 and next 3000 entries of one permutation of 737,280 images are disjoint pools; the 20,000 are divided 90:10 and each index list is

sorted before loading. Images are scaled to $[0, 1]$, with no augmentation. Training seeds are 42, 123, 777, 2024, 31337; the split does not vary with these seeds.

Table 2. Image-data settings used for candidate VAEs. All datasets use 18,000 training, 2000 validation and 3000 test examples. G: Gaussian; B: Bernoulli. Budgets are recorded-time limits with a common limit of 12 candidate-training calls; ID AEs are additional.

Dataset	Input	Likelihood	Candidate widths	Budget (s)
MNIST	$1 \times 28 \times 28$	G	2, 4, 6, 8, 10, 12, 16, 20, 24, 32, 48, 64	1470
Fashion-MNIST	$1 \times 28 \times 28$	G	Same as MNIST	1470
dSprites	$1 \times 64 \times 64$	B	Same as MNIST	5163
CIFAR-10	$3 \times 32 \times 32$	G	8, 16, 24, 32, 48, 64, 96, 128, 192, 256	2924

For MNIST and Fashion-MNIST, the encoder has dense widths 512 and 256, ReLU activations, and separate linear mean/log-variance heads; the decoder has widths 256 and 512 with a linear output. The CIFAR-10 encoder has channels 32, 64 and 128; dSprites adds 256. Convolutions use kernel 4, stride 2, padding 1 and ReLU, leaving 4×4 feature maps, followed by linear posterior heads. A linear/ReLU projection and reversed transposed convolutions form the decoder, ending in a sigmoid. The ID-reference AE flattens the input and uses dense widths $512-256-r$ and the reverse decoder, with a linear output; it is separate from the candidate CNNs.

Ordinary candidates use Adam, learning rate 10^{-3} , default moments $(0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay, batch size 128 and up to 300 epochs. The last batch has 80 examples (141 updates per epoch). ID AEs use the same optimiser and batch size for 100 epochs. Early stopping uses patience 30. A recorded plateau flag requires $(\max h - \min h) / |\text{mean } h| < 0.01$ over the last 50 validation-monitor values. It is not an additional stopping rule. Representative trajectory runs always complete 300 epochs.

The candidate training loss is

$$\mathcal{L}_\beta = \frac{1}{B} \sum_{i=1}^B \left[\ell(x_i, g(z_i)) + \beta \sum_{j=1}^m \frac{\mu_{ij}^2 + \exp(v_{ij}) - 1 - v_{ij}}{2} \right], \quad z_i \sim q_\phi(z | x_i), \quad (3)$$

where v is log variance. Gaussian reconstruction uses a sum of squared errors, equivalent to fixed observation variance $1/2$ up to an additive constant; dSprites uses summed binary cross-entropy. The primary weight is one; the ladder is $\{0.25, 0.5, 1, 2, 4\}$, constant during each run. No cyclical annealing schedule [29] is used. Equal numerical β across likelihoods does not imply equal regularisation strength.

The stored validation monitor named “ELBO” is $h = -[\ell(x, g(\mu(x))) + \text{KL}]$, averaged over examples. It uses posterior-mean reconstruction, not an expectation over posterior samples, and is therefore an *ELBO proxy*, not an unbiased ELBO estimate or a certified bound. It also uses unit KL weight for checkpoint selection even when training uses another β . We retain these checkpoints and label the monitor accurately. No test metric enters the selection code. However, the candidate cache evaluates test metrics for every trained candidate, before method selection; the claim that only one final test evaluation was executed would be incorrect. Tables report test metrics only for the validation-selected models. This is a code-level separation of selection and evaluation, not a prospectively sealed test workflow.

5.2. Methods and Readouts

Grid search selects by validation MSE, the ELBO proxy, or linear-probe validation accuracy. Probes use posterior means, multinomial logistic regression, up to 2000 iterations, and $C \in \{0.01, 0.1, 1, 10\}$ chosen by validation. Labels are digit/apparel/ CIFAR classes

and the three dSprites shapes. Both capacity and probe regularisation use validation labels in the probe selector. Coarse search uses Algorithm 1’s subset; the fixed baseline uses width 32.

FONDUE is a local PyTorch implementation using TwoNN and a fixed ID-gap threshold $0.2d_0$. It searches integer widths, which can lie outside the ordinary candidate grid. Its selected width is trained exactly. At $\beta = 1$, short training uses one epoch for MNIST and dSprites and two for Fashion-MNIST and CIFAR-10. dSprites follows the source’s epoch choice; elsewhere a seed-42 calibration increases epochs until two consecutive predictions agree, stopping at five. Other β -specific choices are in Appendix B. FONDUE+Q maps the returned width upwards onto the grid, obtains the anchor, and extends upwards until quality is met or its budget flag is set. This adds an external reconstruction rule to FONDUE.

FONDUE-VAR starts at $\text{round}(2d_0)$ and doubles width until the declared type condition is met. With mixed variables kept it returns active plus mixed once a passive variable appears; otherwise it returns the active count once a mixed or passive variable appears. We implement the description-based classifier: posterior variance is exponentiated *once* from log variance. A coordinate is passive if its across-data variance is below 0.1 and its mean is in $[0.9, 1.1]$, active if that variance is below 0.1 and its mean is at most 0.1, and mixed otherwise. The audited public-code version is identified in Appendix B.1; this is not claimed to reproduce every path of that code or its published experimental settings.

AU is independently defined by

$$\text{AU}_\delta = \sum_{j=1}^m \mathbf{1}\{\text{Var}_{x \in X_{\text{est}}} \mu_j(x) > \delta\}, \quad \delta \in \{10^{-3}, 10^{-2}, 10^{-1}\}, \quad (4)$$

with primary $\delta = 10^{-2}$. ID and activity readouts use the same 10,000 training examples sampled without replacement using seed 20260915. The MLE cross-check uses $k = 10$ neighbours. AU is diagnostic, not an extra selector in the method comparison.

The GECO and ARD adaptations start at the largest grid width and run 300 epochs. Their raw coordinate counts are mapped to the nearest candidate width, with ties going to the smaller width, for the final ordinary VAE. Appendix B.1 specifies the local objectives, updates and readout differences. The evaluated MSE thus belongs to the transferred ordinary VAE, not to the original model.

5.3. Provenance and Scope of Prior Specification

The saved image-experiment provenance is Python 3.12.3, PyTorch 2.7.1+cu118, CUDA 11.8, cuDNN 9.0.100 (9.1.0), NVIDIA driver 570.211.01, and an NVIDIA GeForce RTX 3070, on Linux 6.8.0-124-generic, x86-64, glibc 2.39. GPU provenance is recorded for these experiments; CPU numerical and probe operations accompany GPU training. The earlier MAKE47 manuscript did state PyTorch 2.0, RTX 3070 and a CPU fallback; it did not identify the complete CUDA/cuDNN/driver environment.

The configuration file was created on 12 September 2026, with subsequent changes recorded on 12–13 September. These are internal specification records, not a public preregistration. Calibration informed some changes, including the dSprites likelihood. The pruning additions and full ladder were run during 14–17 September. The paired random-feature comparison and accounting audit here are retrospective revisions dated 18 September. MNIST and other previously examined datasets are not described as wholly unseen scientific conditions. “Evaluation condition” in the predictive analysis means withheld from fitting that predictor.

The logged code token 8c2c6c28224b3450 hashes only the common module and configuration, not the entire experimental implementation; the earlier trajectory records use 99cb5eb7404ff6e7. A new SHA-256 file manifest identifies the inspected snapshot. Ap-

pendix D maps numerical results to their source files and describes remaining provenance limitations.

6. Results

6.1. Selection Quality and Recorded Cost

Tables 3 and 4 include all four datasets at $\beta = 1$, with mean and sample SD over five candidate-training seeds. They report final model widths, validation and test error, quality attainment and the corrected training-time account. Independent outcome counts appear in Appendix A.1. These uncertainty summaries are conditional on one split and, for ID guidance, one shared reference bank.

Table 3. Dimension selection at $\beta = 1$ (Gaussian likelihoods). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE, including grid rounding for the pruning adaptations. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. C_{train} is the recorded training-time account including the quality anchor, not complete wall-clock cost (Section 3.2).

Method	m_{final}	Val. MSE	Test MSE	Q	C_{train} (s)
<i>MNIST</i>					
Grid (MSE)	12.8 ± 6.4	20.98 ± 0.24	20.50 ± 0.26	5/5	628 ± 35
FONDUE	7.0 ± 0.0	21.93 ± 0.77	21.54 ± 0.78	5/5	104 ± 8
FONDUE + Q	8.0 ± 0.0	21.72 ± 0.50	21.33 ± 0.55	5/5	126 ± 19
GECO adaptation	6.4 ± 0.9	22.40 ± 0.23	22.00 ± 0.24	5/5	278 ± 30
ARD adaptation	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5	238 ± 37
Ascending + Q	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5	182 ± 20
ID-guided + Q	6.0 ± 0.0	22.44 ± 0.15	22.05 ± 0.17	5/5	715 ± 38
<i>Fashion-MNIST</i>					
Grid (MSE)	22.8 ± 15.6	18.33 ± 0.20	18.08 ± 0.19	5/5	645 ± 42
FONDUE	6.8 ± 0.4	19.29 ± 0.16	19.01 ± 0.16	5/5	100 ± 10
FONDUE + Q	7.6 ± 0.9	19.00 ± 0.33	18.75 ± 0.34	5/5	108 ± 15
GECO adaptation	4.0 ± 0.0	19.38 ± 0.24	19.06 ± 0.25	5/5	349 ± 27
ARD adaptation	3.6 ± 0.9	20.67 ± 2.75	20.36 ± 2.80	4/5	261 ± 19
Ascending + Q	4.0 ± 0.0	19.38 ± 0.24	19.06 ± 0.25	5/5	209 ± 26
ID-guided + Q	6.4 ± 3.3	19.03 ± 0.61	18.73 ± 0.57	5/5	457 ± 32

Table 4. Dimension selection at $\beta = 1$ (Bernoulli for dSprites; Gaussian for CIFAR-10). Values are mean $\pm$ sample SD over five candidate-training seeds. m_{final} is the width of the evaluated ordinary VAE, including grid rounding for the pruning adaptations. MSE columns are multiplied by 10^3 ; Q counts validation target attainment. C_{train} is the recorded training-time account including the quality anchor, not complete wall-clock cost (Section 3.2).

Method	m_{final}	Val. MSE	Test MSE	Q	C_{train} (s)
<i>dSprites</i>					
Grid (MSE)	57.6 ± 14.3	1.16 ± 0.01	1.15 ± 0.01	5/5	2565 ± 82
FONDUE	4.2 ± 0.4	5.50 ± 1.31	5.42 ± 1.28	0/5	378 ± 25
FONDUE + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5	1676 ± 224
GECO adaptation	64.0 ± 0.0	1.17 ± 0.02	1.16 ± 0.03	5/5	969 ± 10
ARD adaptation	8.4 ± 1.7	1.92 ± 0.60	1.88 ± 0.60	0/5	1029 ± 32
Ascending + Q	20.0 ± 4.0	1.26 ± 0.05	1.24 ± 0.05	5/5	1797 ± 266
ID-guided + Q	64.0 ± 0.0	1.17 ± 0.02	1.16 ± 0.03	5/5	1001 ± 38
<i>CIFAR-10</i>					
Grid (MSE)	140.8 ± 53.5	13.83 ± 0.04	14.07 ± 0.04	5/5	428 ± 16
FONDUE	14.4 ± 0.5	17.79 ± 0.30	18.10 ± 0.30	0/5	110 ± 16
FONDUE + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5	133 ± 17
GECO adaptation	19.2 ± 4.4	16.29 ± 1.18	16.57 ± 1.20	2/5	452 ± 14
ARD adaptation	16.0 ± 0.0	17.09 ± 0.09	17.39 ± 0.08	0/5	366 ± 12
Ascending + Q	24.0 ± 0.0	15.12 ± 0.14	15.36 ± 0.13	5/5	157 ± 20
ID-guided + Q	131.2 ± 113.9	14.01 ± 0.13	14.25 ± 0.14	5/5	499 ± 18

On MNIST, ID guidance and ascending search return $m = 6$ in all five trials and share the same final models. Their validation MSE is 0.02244; all meet the target. The recorded training accounts are 715 and 182 s, respectively (a ratio of approximately 3.9). The difference mainly reflects the reference-AE bank and additional candidates (Table 5); its missing ID-estimation overhead cannot support a more favourable ID cost. It remains a comparison of the executed procedures rather than proof that every possible ID-guided search is slower.

On dSprites, ordinary FONDUE returns approximately four dimensions but has validation MSE 0.00550, versus 0.00126 for ascending search. The target column shows whether this faster selection reaches comparable quality. Normal termination alone must not be read as quality success. The FONDUE+Q dSprites result includes one budget-exceeded trial with an existing returned model; it is retained in the means.

The ID check fails on Fashion-MNIST, dSprites and CIFAR-10, so those rows are coarse-search fallbacks with the reference cost added. They are not examples of successful ID placement. The high variability on CIFAR-10 belongs to this fallback. dSprites is a synthetic image dataset; the only natural colour-image benchmark here is CIFAR-10.

Table 5. MNIST training-time decomposition at $\beta = 1$ (mean seconds, five seeds). The final model is charged once and removed from the other-candidate column. ID computation and post-training evaluation were not separately timed. FONDUE epoch calibration adds 4.14 s once per dataset- β setting and is not included below.

Method	Anchor	ID AEs	Other training	Final	Total
Grid (MSE)	69.3	0.0	502.0	56.2	627.5
FONDUE	62.6	0.0	1.4	39.9	103.9
FONDUE + Q	69.3	0.0	1.4	55.3	125.9
GECO adaptation	62.6	0.0	175.4	39.6	277.6
ARD adaptation	62.6	0.0	138.0	37.6	238.2
Ascending + Q	69.3	0.0	72.8	39.9	182.0
ID-guided + Q	69.3	205.5	400.7	39.9	715.4

6.2. Representation and Sample Dependence of ID

Table 6 gives the complete reference-bank summary, including Fashion-MNIST. Only MNIST passes both checks. Because the same bank is reused, five identical pass/fail decisions do not constitute five independent replications of ID estimation.

Table 6. ID-reference AE widths and TwoNN estimates (mean $\pm$ SD over three reference seeds); stability R and estimator disagreement A . Limits are $R \leq 0.15$ and $A \leq 0.25$. The same reference bank is reused across candidate seeds and β .

Dataset	Width: estimate	R	A
MNIST	55: 10.93 $\pm$ 0.08; 111: 11.78 $\pm$ 0.12; 221: 12.32 $\pm$ 0.09	0.118	0.150
Fashion-MNIST	49: 9.91 $\pm$ 0.06; 98: 10.45 $\pm$ 0.04; 197: 10.83 $\pm$ 0.05	0.088	0.361
dSprites	30: 3.96 $\pm$ 0.48; 61: 3.91 $\pm$ 1.12; 122: 2.10 $\pm$ 0.02	0.474	1.816
CIFAR-10	81: 13.60 $\pm$ 0.09; 162: 13.44 $\pm$ 0.71; 325: 12.34 $\pm$ 0.50	0.094	0.834

Figure 2 separately shows the sample-size and PCA analyses. At 1000 examples the two estimators agree within the specified 0.25 relative tolerance on all four raw-input datasets; at 10,000, only MNIST does. These are raw-input checks, distinct from the AE-latent checks used by the selector. On dSprites, input TwoNN is 7.70 in this diagnostic, PCA estimates range from 2.68 to 4.32, and individual AE estimates range from 2.08 to 4.56. The combined range is about 3.7-fold; the corresponding MNIST range is about 1.7-fold. These are representation-dependent estimates, not competing measurements of a known ground truth.

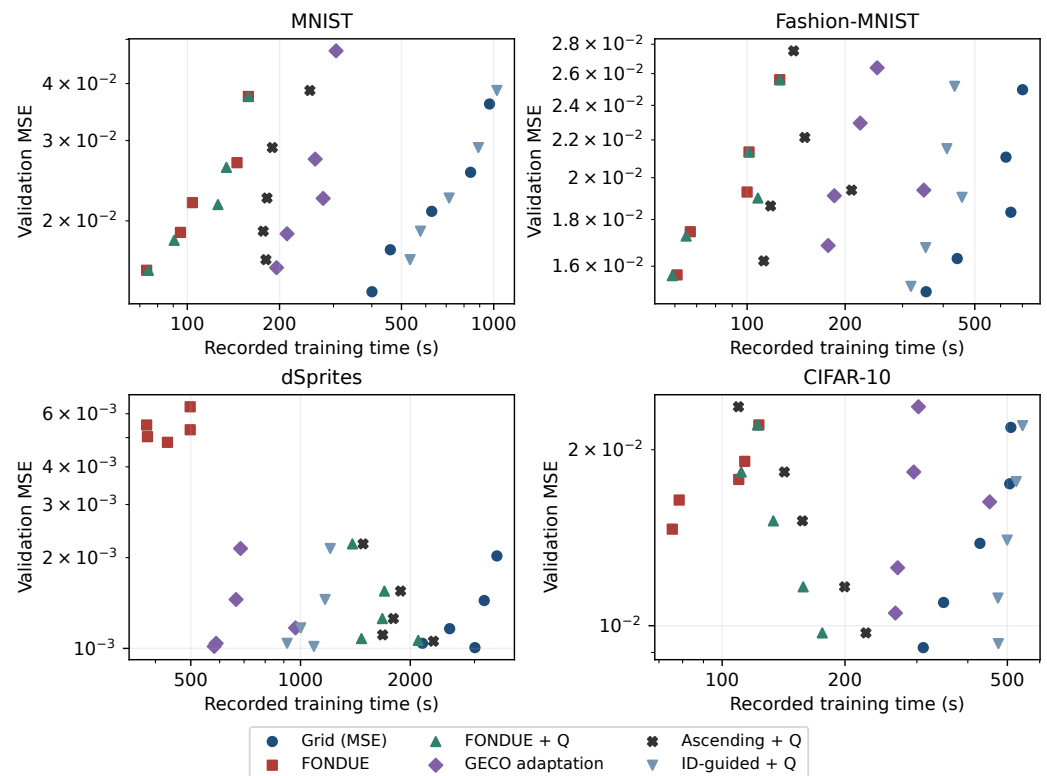


Figure 1. Validation quality versus attributed recorded training time, five seed means for each of five regularisation settings. Gaussian likelihood except Bernoulli dSprites. ARD is omitted because its selection stage did not vary with β . GECCO changes the anchor-derived constraint. These points do not all meet a common fixed-quality target.

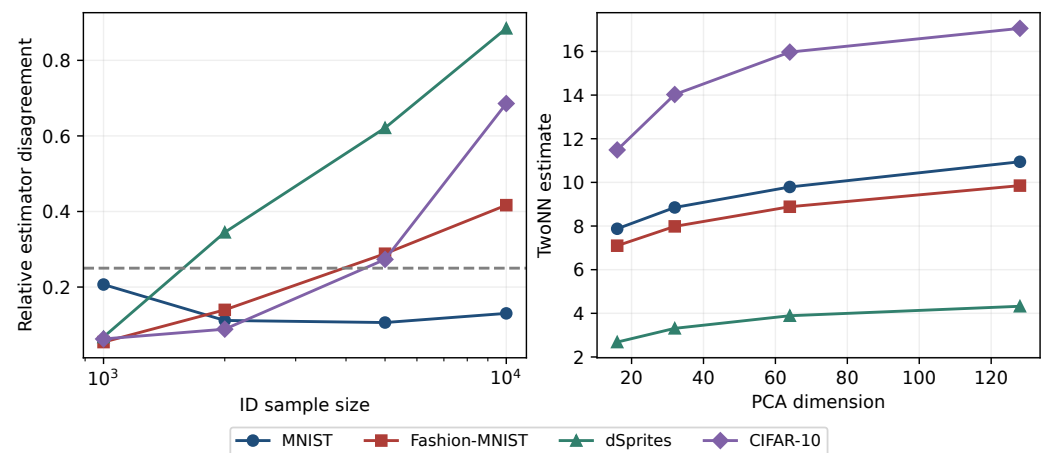


Figure 2. Left: $|d_{\text{TwoNN}} - d_{\text{MLE}}|/d_{\text{TwoNN}}$ on raw inputs as estimation sample size changes; dashed line is 0.25. Right: TwoNN after PCA preprocessing. PCA rank, sample size and representation are distinct interventions; AE results are in Table 6.

6.3. Dependence on Regularisation and Quality Context

Define the reported range for method a and dataset d by

$$S_{a,d} = \max_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta) - \min_{\beta \in \mathcal{B}} \bar{m}_{a,d}(\beta), \quad \bar{m}_{a,d}(\beta) = \frac{1}{5} \sum_{s=1}^5 m_{a,d,s}(\beta), \quad \mathcal{B} = \{0.25, 0.5, 1, 2, 4\}. \quad (5)$$

This is the range of seed means of raw returned dimensions, not the mean of within-seed ranges. For the FONDUE-VAR row all five outputs are defined at each setting, including any raw zero. It is a descriptive range on the tested ladder, not a new parameter-free robustness index.

Table 7. Range of seed-mean returned dimensions across $\beta \in \{0.25, 0.5, 1, 2, 4\}$. GECO varies the anchor-derived constraint; it is not a direct β intervention. ARD selection was fixed at an implicit weight of one and is excluded. Fixed 32 is a design constant; fallback spans are not ID robustness. Gaussian likelihood except Bernoulli dSprites.

Method	MNIST	Fashion	dSprites	CIFAR-10
Grid (MSE)	14.8	20.8	40.0	91.2
Coarse search	33.6	37.2	0.0	214.4
Fixed 32	0.0	0.0	0.0	0.0
FONDUE	3.8	0.8	0.8	8.8
FONDUE-VAR (mixed kept)	50.6	63.0	106.2	68.8
Ascending + Q	6.0	3.6	15.2	52.8
FONDUE + Q	3.6	1.6	28.0	44.8
ID-guided + Q	6.0	37.2	0.0	214.4
GECO adaptation	8.8	2.0	0.0	34.2
ARD adaptation	—	—	—	—

Ranges differ by method and dataset, with clear ordering reversals. For example, ID-guided versus GECO-adaptation ranges are 6.0 versus 8.8 on MNIST, but 214.4 versus 34.2 on CIFAR-10. The latter ID-guided value is inherited from fallback. A fixed dimension has zero range by design; dSprites ID fallback has zero range because coarse search always returns 64. Neither demonstrates useful selection robustness. We do not average ranges across differently scaled grids or assert a transferable ranking.

The ARD call contains no selection-stage β argument and optimises reconstruction plus KL with unit weight every time. Its five contextual runs repeat the same selection settings and seeds. The final ordinary VAE changes with the contextual β , but the ARD-selected count does not. The former zero-range row is therefore excluded from sensitivity comparison. GECO has a different intervention: its constraint is $\tau = T_{D,s}(\beta)$, while its KL coefficient remains one. These distinctions are explicit in Figure 3.

An AU-threshold intervention holds the checkpoint fixed and recomputes Equation (4). A quality-margin intervention instead changes which cached models satisfy the target. Neither is the same as changing training β . We report these readout/quality analyses separately (Section 6.5 and Appendix B.2). No sensitivity claim is made for untested FONDUE stopping thresholds or ARD cumulative relevance fractions.

6.4. Learning Curves, Test Evaluation and Selection Criteria

The saved trajectory experiment supplies train/validation curves for MNIST widths 12, 32 and 64 and dSprites widths 6, 16 and 32, with five seeds each (Figure 4). Training MSE is evaluated in deterministic mode on a fixed 2000-example training subset sampled with seed 20260919. Both train and validation improve, but validation error remains higher. None of these 30 runs passes the stringent final 50-epoch plateau check; completing 300

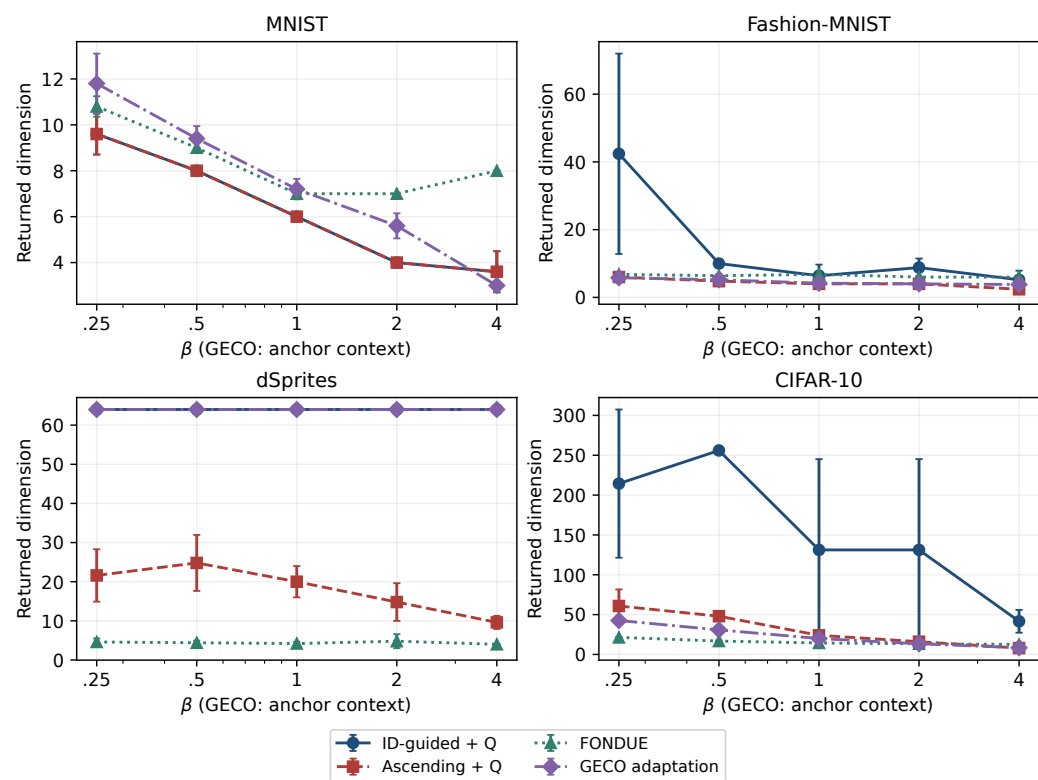


Figure 3. Raw returned dimension across the five contexts, mean $\pm$ SD over five candidate seeds. Gaussian likelihood except Bernoulli dSprites. Ordinary VAE selectors change training β ; GECO changes its anchor-derived constraint. ARD is excluded. ID guidance falls back on Fashion-MNIST, dSprites and CIFAR-10 throughout the ladder.

epochs is not proof of convergence. These curves cover representative settings, not all four datasets.

Table 8. Selection-criterion comparison at $\beta = 1$, five seeds; test MSE $\times 10^3$ and probe test accuracy (%). Probes use class labels (shape for dSprites); only Grid (probe) uses them for capacity selection. Recorded probe time is charged to that selector. Other displayed probe accuracies are auxiliary evaluations with uncharged costs. Likelihoods follow Table 2.

Dataset	Criterion	m	Test MSE	Accuracy	Time (s)
MNIST	Grid (MSE)	12.8 ± 6.4	20.50 ± 0.26	89.4 ± 0.9	628 ± 35
MNIST	Grid (ELBO proxy)	10.0 ± 2.0	20.54 ± 0.25	89.3 ± 0.7	628 ± 35
MNIST	Grid (probe)	51.2 ± 13.4	23.01 ± 0.55	92.7 ± 0.5	696 ± 36
Fashion-MNIST	Grid (MSE)	22.8 ± 15.6	18.08 ± 0.19	75.5 ± 2.6	645 ± 42
Fashion-MNIST	Grid (ELBO proxy)	15.6 ± 8.0	18.12 ± 0.27	74.4 ± 2.6	645 ± 42
Fashion-MNIST	Grid (probe)	57.6 ± 8.8	18.99 ± 0.46	78.4 ± 0.6	750 ± 40
dSprites	Grid (MSE)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (ELBO proxy)	57.6 ± 14.3	1.15 ± 0.01	47.7 ± 2.3	2565 ± 82
dSprites	Grid (probe)	64.0 ± 0.0	1.16 ± 0.03	49.8 ± 3.0	2589 ± 84
CIFAR-10	Grid (MSE)	140.8 ± 53.5	14.07 ± 0.04	44.3 ± 2.6	428 ± 16
CIFAR-10	Grid (ELBO proxy)	192.0 ± 45.3	14.11 ± 0.08	46.1 ± 1.1	428 ± 16
CIFAR-10	Grid (probe)	256.0 ± 0.0	14.30 ± 0.12	47.3 ± 1.0	574 ± 15

Table 8 provides the previously omitted grid choices based on MSE, the ELBO proxy and a linear probe. Different objectives can select different widths and reconstruction errors. Test probe accuracy is an auxiliary supervised outcome; it does not measure unsupervised reconstruction quality. This comparison retains the same splits and cached candidates, with labelled validation selection explicitly stated.

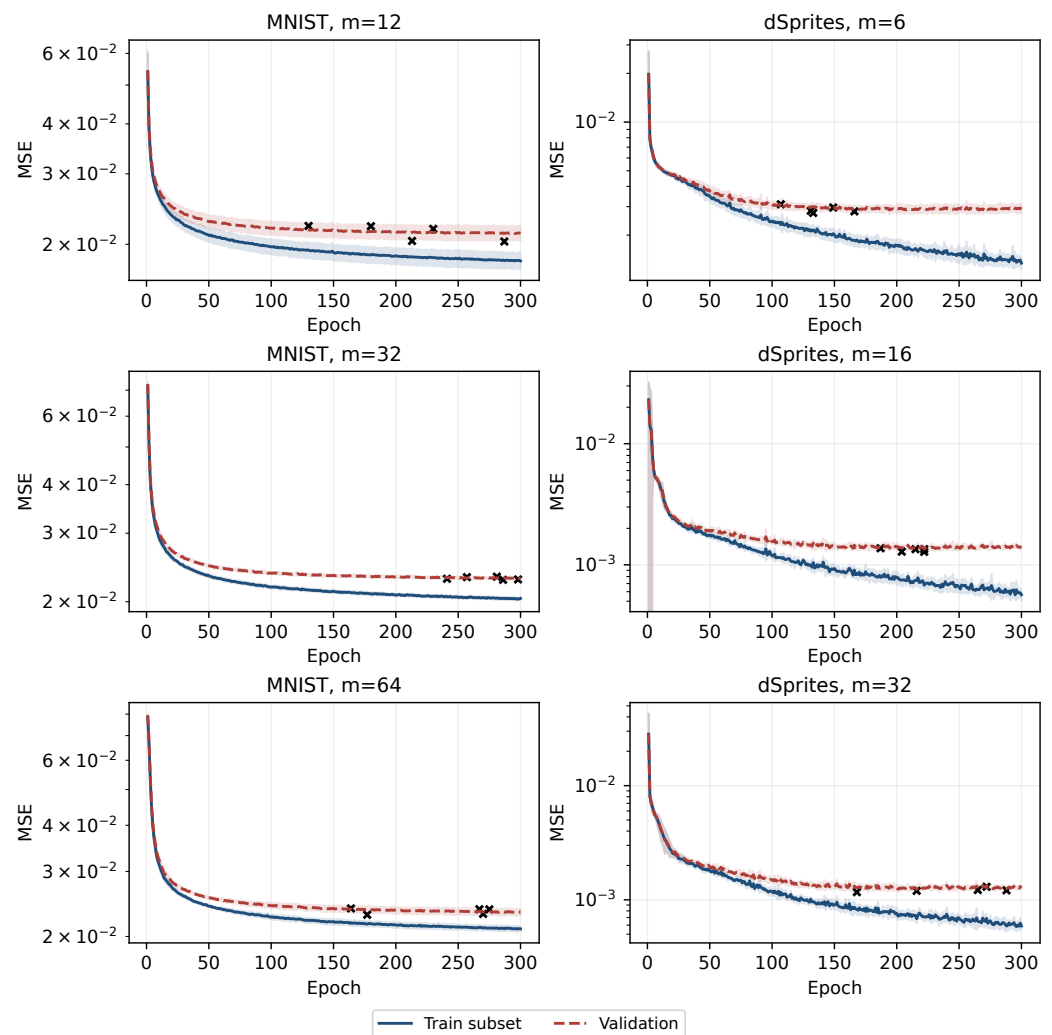


Figure 4. Train and validation MSE, means with ± 1 SD bands over five seeds. MNIST uses Gaussian and dSprites Bernoulli likelihood; $\beta = 1$. Black crosses mark each run's checkpoint selected by the validation ELBO proxy. These diagnostic runs have no early stopping.

6.5. Activity Readout: Trajectories, Thresholds and Prediction

Figure 5 and Table 9 compare the two counts on identical checkpoints. At MNIST width 64, epoch-five variable types give 9.8 ± 3.8 and AU gives 49.4 ± 8.1 ; at epoch 300 both give 6.4 ± 0.5 . This example distinguishes mean agreement from between-seed variability. The table also reports within-run changes from 50 to 300 epochs. Their magnitude depends on the condition, so no general stability superiority is inferred from the final agreement.

Table 9. Variable-type count V (active plus mixed) and AU count A , five seeds, $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Mean $\pm$ SD quantify between-seed variation. The last column gives mean absolute within-run changes from epoch 50 to 300 for V/A , not a temporal variance.

Dataset	m	V_5	A_5	V_{300}	A_{300}	Change
MNIST	12	7.2 ± 0.8	8.0 ± 1.6	6.4 ± 0.5	6.4 ± 0.5	0.0/0.0
MNIST	32	8.2 ± 1.5	20.8 ± 4.2	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
MNIST	64	9.8 ± 3.8	49.4 ± 8.1	6.4 ± 0.5	6.4 ± 0.5	0.4/2.6
dSprites	6	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	0.0/0.0
dSprites	16	16.0 ± 0.0	16.0 ± 0.0	14.2 ± 0.4	13.4 ± 0.5	1.8/1.8
dSprites	32	32.0 ± 0.0	31.8 ± 0.4	14.2 ± 1.1	14.4 ± 1.1	12.4/4.2

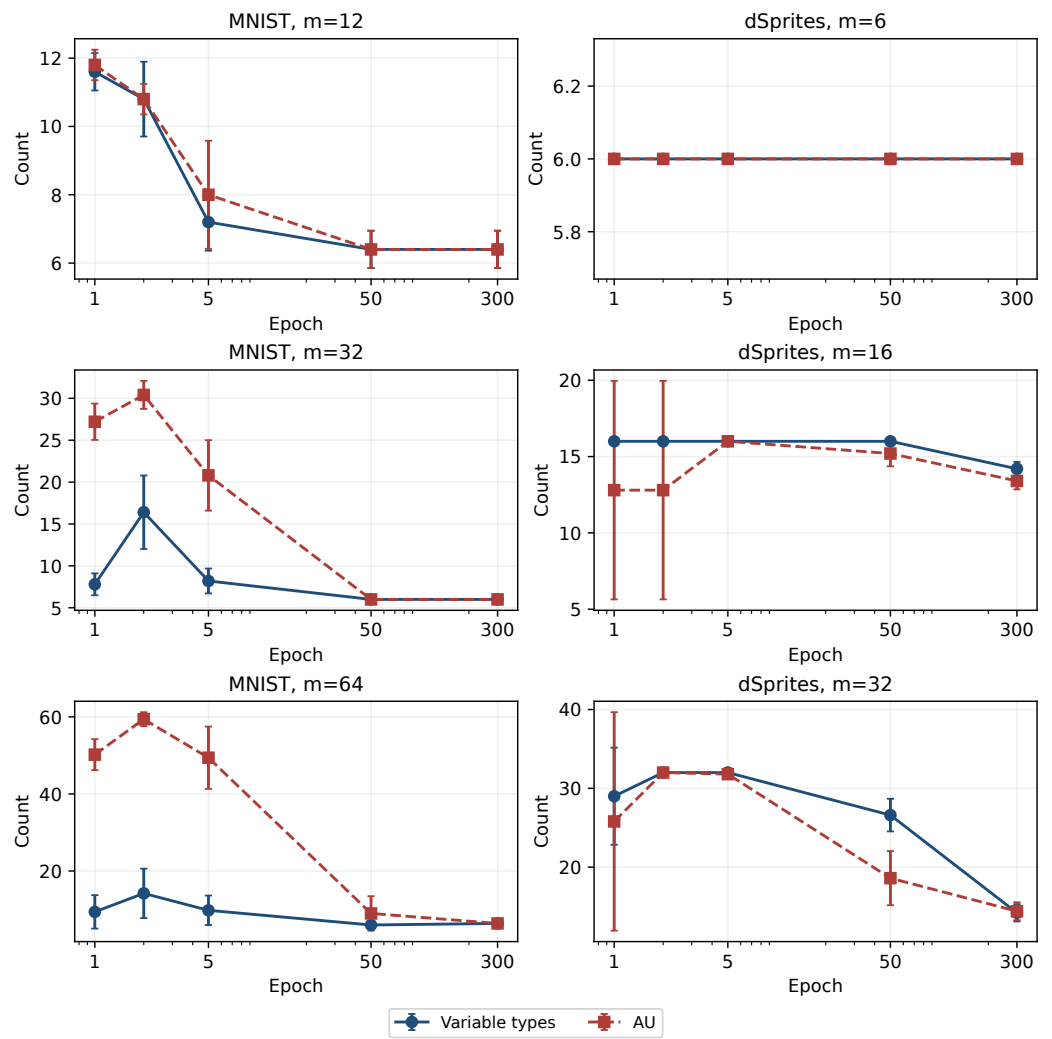


Figure 5. Variable-type count (active plus mixed) and AU from the same checkpoints at epochs 1, 2, 5, 50 and 300; means and seed SDs, with lines connecting the observed checkpoints only. $\beta = 1$, Gaussian MNIST and Bernoulli dSprites. Constant full-capacity counts are not evidence of diagnostic stability.

The threshold-only comparison in Figure 6 uses fixed $\beta = 1$ checkpoints. On CIFAR-10, $\delta = 0.001$ counts every coordinate at every tested grid width, whereas $\delta = 0.1$ counts far fewer at large widths. The inference “no unused coordinates” therefore depends on this threshold. Normalising the threshold changes the diagnostic definition; it does not establish a dimension estimate.

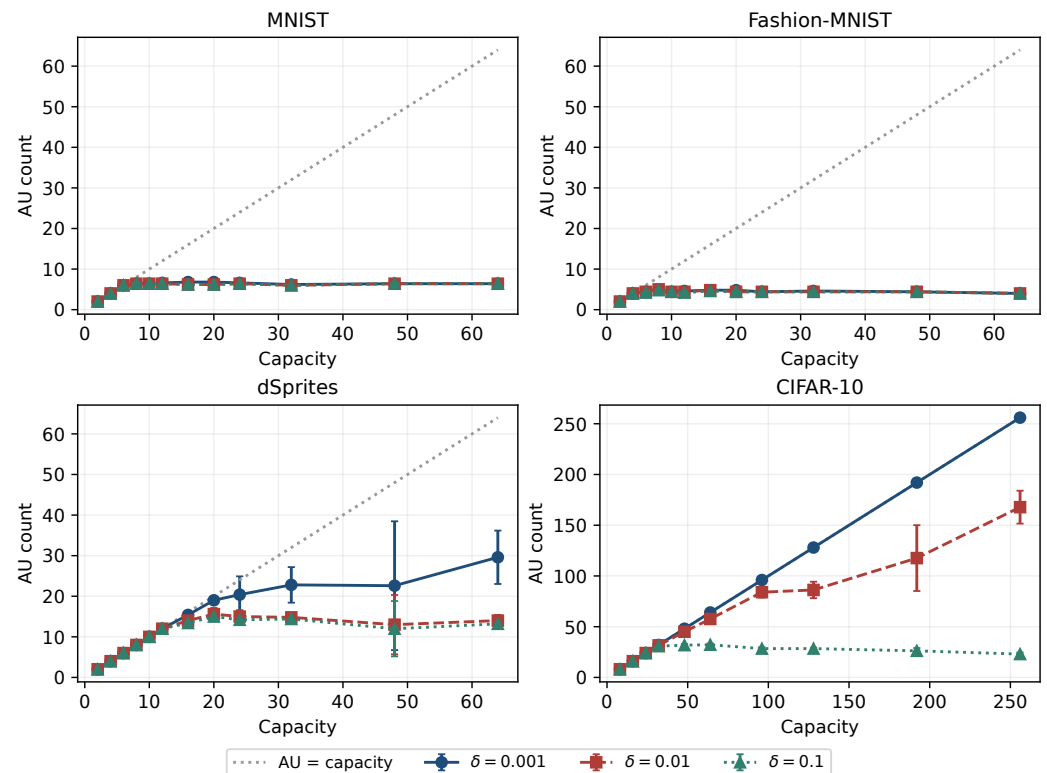


Figure 6. AU under three thresholds on the same $\beta = 1$ candidates; mean $\pm$ seed SD. Gaussian likelihood except Bernoulli dSprites. The diagonal is capacity. This is a readout-threshold intervention, unlike the training/context intervention in Figure 3.

The predictive task asks whether the next smaller grid candidate meets the same seed’s quality target. Each row is one validation-selected checkpoint at a nonminimal grid width. Baseline features are validation MSE, KL, the ELBO proxy, width, width/input-ID ratio, and the slope of validation MSE over the last 20 recorded epochs. The three added AU features are AU, AU/width and AU/input-ID. Standardisation is fitted only on development rows. An L_2 logistic predictor is fitted on MNIST (55 rows), with $C = 10$ chosen on a three-seed/two-seed development split. Evaluation uses dSprites (55 rows), Fashion-MNIST (55) and CIFAR-10 (45), each with five seeds. The configured MNIST-CNN predictive condition was not executed.

The development split uses seeds 42, 123 and 777 for the inner fit and 2024 and 31337 for selecting C , then refits on all five. Predictors use up to 5000 iterations. Each evaluation dataset–seed block is scored over its complete set of nonminimal grid widths; checkpoint rows are not resampled as independent observations.

The original internally specified analysis reported an AU-minus-baseline AU-ROC improvement of $+0.108$, with a 2000-replicate seed-record bootstrap 95% interval $[+0.059, +0.157]$. A subsequently added control used three independent standard-normal features and one noise realisation (generator seed 7); its separately reported improvement was $+0.098$ with interval $[-0.025, +0.227]$. Overlap of these intervals does not test their difference or establish equivalence. These archived values are retained as the original

analysis, not as a validated positive finding: its AUROC calculation used probabilities that can saturate numerically under the shift between datasets.

Table 10. Retrospective paired AU comparison at $\beta = 1$, using logistic decision scores to avoid saturated-probability ties. Values are means of five seed-specific AUROCs, each evaluated over all nonminimal capacities in that seed. Random features are three standard-normal columns with generator seed 7. All methods use the same rows and labels.

Evaluation condition	Rows	Base	Base + AU	Base + random	AU – random
dSprites	55	0.950	0.858	0.898	-0.041
Fashion-MNIST	55	0.980	1.000	0.880	+0.120
CIFAR-10	45	1.000	1.000	1.000	+0.000

For this revision we reconstruct both predictors at explicitly fixed $\beta = 1$, restrict rows to the declared grid, and compute AUROC from logistic decision scores. The original cache-reading routine did not filter β or restrict the grid, so rerunning it after cache expansion would also be ambiguous. Decision scores preserve ranks that are lost when sigmoid probabilities round to exactly one. Saturation counts appear in Appendix B.2. This is a corrected retrospective analysis, not an exact replay of archived probabilities. The AU-minus-baseline difference is now -0.024 , with interval $[-0.102, +0.027]$. We also compare AU and random predictors on the same dataset-seed blocks. The paired mean difference is $+0.026$, with a retrospective hierarchical-bootstrap 95% interval $[-0.040, +0.133]$ (2000 replicates). Each replicate resamples the three evaluation datasets and then the five seed blocks within each selected dataset; all capacities and both predictors remain paired within a block. Models are held fixed, so this interval does not cover development-model fitting uncertainty. Only three evaluation datasets are available, limiting precision of across-condition inference. A secondary analysis averages the comparator over 100 noise realisations (seeds 0–99), giving a paired difference -0.007 and interval $[-0.064, +0.039]$ under the same resampling. Neither analysis establishes an AU-specific benefit in these evaluation conditions. We do not claim that AU has no information, that the predictors are equivalent, or that lack of significance proves insufficient power.

6.6. Implementation Audit and Degenerate Outputs

Two scale inconsistencies were identified in local implementations. First, the earlier VAE loss averaged KL over latent dimensions while summing reconstruction over input dimensions. Algebraically this gives $\beta_{\text{effective}} = \beta_{\text{nominal}}/m$. The image results in this paper come from the corrected sum-over-latents convention. The algebra does not imply that complete optimisation trajectories or AU counts are reproduced simply by relabelling β ; the earlier “within 1.7 units” claim is not used here.

Second, an intermediate GECCO adaptation combined per-pixel-mean constraint values with other terms on incompatible scales. The corrected implementation uses summed squared error and multiplies the per-pixel target by the input dimension. Either sum or mean conventions can be valid if the objective, target and multiplier scales are consistent. The observed behaviour of this hard-concrete adaptation is not attributed to the original ARM implementation.

FONDUE-VAR’s active-only setting returning zero is a separate phenomenon, not a third sum/mean defect. At MNIST’s one-epoch setting it returns zero on all five trials; no final VAE exists for these outputs. Table 11 shows both settings and their quality/validity counts.

Table 11. FONDUE-VAR at $\beta = 1$: both keep-mixed settings, valid-model and quality counts, test MSE $\times 10^3$, and recorded training time. Raw zero outputs remain in the dimension summary; nonexistent models have no MSE. Five seeds; likelihoods follow Table 2.

Dataset	Mixed kept	Raw m	Valid	Q	Test MSE	Time (s)
MNIST	yes	9.8 ± 8.5	5/5	5/5	22.53 ± 0.89	113 ± 22
MNIST	no	0.0 ± 0.0	0/5	0/5	—	63 ± 10
Fashion-MNIST	yes	14.4 ± 3.8	5/5	5/5	18.96 ± 0.39	91 ± 11
Fashion-MNIST	no	2.0 ± 0.0	5/5	0/5	25.27 ± 0.06	120 ± 18
dSprites	yes	26.0 ± 49.5	4/5	1/5	9.53 ± 9.62	412 ± 118
dSprites	no	1.2 ± 1.6	2/5	0/5	7.71 ± 0.02	278 ± 49
CIFAR-10	yes	34.2 ± 1.3	5/5	5/5	14.40 ± 0.15	103 ± 13
CIFAR-10	no	3.2 ± 0.8	5/5	0/5	31.65 ± 2.68	96 ± 13

7. Discussion

The most direct evidence is the paired MNIST comparison: under the tested model family, grid and target, the implemented ID-guided selector gives no recorded-training-cost advantage over ascending search. Its reliability checks reject the other three datasets, but those repeated rejections share a reference bank. They indicate failure of this applicability check under these representations, not that ID methods generally cannot work on those datasets. Different reference architectures, grids, targets or reliable estimators could change the comparison.

Regularisation ranges and AU-threshold responses are different measurements. Some large dimension ranges reflect fallback, grid scale or a changing quality target. A constant output can be fixed by design or by an unvaried implementation argument. Thus the evidence supports method- and dataset-dependent responses, not a universal robustness ranking. The archived positive AU-versus-baseline result is not robust to preserving logistic-score rankings instead of rounded probabilities. The corrected predictive comparison does not establish an AU-specific advantage over the matched random-feature controls.

Several limits remain material. The pruning rows evaluate dimension transfer through local adaptations, including nearest-grid rounding; their native model performance and exact-source implementations are not compared. Complete ID/evaluation timings are missing, so a strict end-to-end cost ranking cannot be reconstructed. The ID selector’s expansion branch is ineffective. A fixed-quality β intervention, a correctly varied ARD selection weight with source-matched relevance scoring, and an exact ARM comparison require new training and are not claimed as completed experiments. The single split and shared references also limit uncertainty estimates. These restrictions are preferable to interpreting the current records as broader evidence than they provide.

The synthetic experiments in Appendix C illustrate scale dependence and the effect of zeroing low-variance coordinates. Their counts are observations under a particular loss and coordinate system. They do not show that topology predicts the number of active axes, and their preprocessing prevents treating them as clean estimates of held-out generalisation.

8. Conclusions

We compared latent-width selection rules using common candidate-model families, explicit validation targets and a documented training-time account. At the primary MNIST setting, ID guidance and ascending search select the same six-dimensional VAE, while ID guidance incurs more recorded training time. ID estimates depend on representation and sample size; AU counts depend on their readout threshold. The paired predictive comparison does not establish an AU-specific advantage over the evaluated random-feature controls. Dimension variation across regularisation contexts differs by method and dataset, without a transferable ranking. These findings delimit what ID and AU alone can establish about the capacity needed for a chosen reconstruction quality under the tested conditions.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Quality Targets and Outcome Accounting

Table A1. Quality anchors and targets, in units of 10^{-3} MSE, mean $\pm$ SD over five seeds. Each decision uses its own seed-specific target, not the displayed mean. The relative margin is 10%; the absolute floor is inactive. Gaussian likelihood except Bernoulli dSprites.

Dataset	β	Anchor MSE	Target $T_{D,s}$
MNIST	0.25	15.843 $\pm$ 0.562	17.428 $\pm$ 0.618
MNIST	0.5	19.202 $\pm$ 0.497	21.122 $\pm$ 0.546
MNIST	1	23.712 $\pm$ 0.532	26.084 $\pm$ 0.585
MNIST	2	30.061 $\pm$ 0.560	33.068 $\pm$ 0.617
MNIST	4	38.977 $\pm$ 0.673	42.875 $\pm$ 0.740
Fashion-MNIST	0.25	15.591 $\pm$ 0.678	17.150 $\pm$ 0.746
Fashion-MNIST	0.5	17.408 $\pm$ 0.513	19.148 $\pm$ 0.564
Fashion-MNIST	1	19.679 $\pm$ 0.162	21.647 $\pm$ 0.179
Fashion-MNIST	2	22.416 $\pm$ 0.527	24.658 $\pm$ 0.580
Fashion-MNIST	4	25.967 $\pm$ 0.587	28.564 $\pm$ 0.645
dSprites	0.25	1.015 $\pm$ 0.032	1.117 $\pm$ 0.036
dSprites	0.5	1.040 $\pm$ 0.032	1.144 $\pm$ 0.035
dSprites	1	1.170 $\pm$ 0.024	1.287 $\pm$ 0.026
dSprites	2	1.453 $\pm$ 0.038	1.598 $\pm$ 0.041
dSprites	4	2.144 $\pm$ 0.064	2.358 $\pm$ 0.070
CIFAR-10	0.25	9.407 $\pm$ 0.610	10.348 $\pm$ 0.671
CIFAR-10	0.5	11.151 $\pm$ 0.200	12.266 $\pm$ 0.220
CIFAR-10	1	14.064 $\pm$ 0.108	15.470 $\pm$ 0.119
CIFAR-10	2	17.806 $\pm$ 0.262	19.587 $\pm$ 0.288
CIFAR-10	4	22.399 $\pm$ 0.286	24.638 $\pm$ 0.314

Appendix A.1. Independent Outcome Counts

The following tables separate the original combined state from flags and final-model quality. A budget flag can coexist with ID fallback. The pruning wrapper did not enforce the ordinary selector budget during training, so its zero budget-flag count is not evidence of equal budget control. All valid returned models, including those from budget-exceeded

trials, contribute to the primary means. Degenerate outputs contribute to raw-count and cost summaries but not to MSE.

Table A2. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
MNIST	Grid (MSE)	S5	0	0	5	5
MNIST	FONDUE	S5	0	0	5	5
MNIST	FONDUE + Q	S5	0	0	5	5
MNIST	GECO adaptation	S5	0	0	5	5
MNIST	ARD adaptation	S5	0	0	5	5
MNIST	Ascending + Q	S5	0	0	5	5
MNIST	ID-guided + Q	S5	0	0	5	5
MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
MNIST	FONDUE-VAR (active only)	D5	0	0	0	0
Fashion-MNIST	Grid (MSE)	S5	0	0	5	5
Fashion-MNIST	FONDUE	S5	0	0	5	5
Fashion-MNIST	FONDUE + Q	S5	0	0	5	5
Fashion-MNIST	GECO adaptation	S5	0	0	5	5
Fashion-MNIST	ARD adaptation	S5	0	0	4	5
Fashion-MNIST	Ascending + Q	S5	0	0	5	5
Fashion-MNIST	ID-guided + Q	I5	5	0	5	5
Fashion-MNIST	FONDUE-VAR (mixed kept)	S5	0	0	5	5
Fashion-MNIST	FONDUE-VAR (active only)	S5	0	0	0	5

Table A3. Outcome audit at $\beta = 1$; all counts have denominator five. S: recorded success; I: ID unreliable; B: budget exceeded; D: degenerate; N: no compact alternative. FB is the fallback flag; budget, validation quality Q , and compression of the final VAE are independently recomputed or read from flags.

Dataset	Method	State	FB	Budget	Q	Compact
dSprites	Grid (MSE)	S5	0	0	5	1
dSprites	FONDUE	S5	0	0	0	5
dSprites	FONDUE + Q	B1, S4	0	1	5	5
dSprites	GECO adaptation	S5	0	0	5	0
dSprites	ARD adaptation	S5	0	0	0	5
dSprites	Ascending + Q	S5	0	0	5	5
dSprites	ID-guided + Q	I5	5	0	5	0
dSprites	FONDUE-VAR (mixed kept)	D1, S4	0	0	1	3
dSprites	FONDUE-VAR (active only)	D3, S2	0	0	0	2
CIFAR-10	Grid (MSE)	S5	0	0	5	5
CIFAR-10	FONDUE	S5	0	0	0	5
CIFAR-10	FONDUE + Q	S5	0	0	5	5
CIFAR-10	GECO adaptation	S5	0	0	2	5
CIFAR-10	ARD adaptation	S5	0	0	0	5
CIFAR-10	Ascending + Q	S5	0	0	5	5
CIFAR-10	ID-guided + Q	I5	5	0	5	3
CIFAR-10	FONDUE-VAR (mixed kept)	S5	0	0	5	5
CIFAR-10	FONDUE-VAR (active only)	S5	0	0	0	5

Appendix B. Coverage and Implementation Details

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Table A4. Executed coverage. Each image method row has five training seeds at each listed context. Additional diagnostic experiments have their own seed units.

Experiment	Dataset coverage	Contexts / counts
Fourteen ordinary selectors/controls	All four image datasets	Five β values; $14 \times 4 \times 5 \times 5 = 1400$ method records
GECO adaptation	All four	Five anchor-derived constraints; 100 records
ARD adaptation	All four	Fixed selection-stage weight one; five repeated contexts, 100 records
ID-reference bank	All four	Three widths $\times$ three seeds each, shared across contexts
Train/validation and readout trajectories	MNIST: 12, 32, 64; dSprites: 6, 16, 32	$\beta = 1$, five seeds each; 30 runs
AU prediction	Develop: MNIST; evaluate: dSprites, Fashion-MNIST, CIFAR-10	$\beta = 1$; 55 development and 155 evaluation rows
Synthetic diagnostic	Swiss roll and torus	Six capacities, three seeds, two scaling options; 72 runs

The fourteen ordinary method records comprise MSE/ELBO-proxy/probe grid selection, coarse search, fixed width, FONDUE, two FONDUE-VAR settings, ID guidance with and without AU recording, ascending+Q, and coarse, FONDUE and FONDUE-VAR with added Q. Removing AU recording does not change the ID selection decisions; its overhead was not separately timed. Machine-readable results include all fourteen, even when only the principal comparisons are printed.

Table A5. Split identities reconstructed from the saved loader and fixed seeds. Tokens are the first 16 hexadecimal characters of SHA-256 over concatenated int64 train/validation/test index arrays, matching the loader convention. Official training/test pools have separate index spaces.

Dataset	Reconstructed split token
MNIST	cffaecf5bccbb333
Fashion-MNIST	cffaecf5bccbb333
dSprites	1ea516804f7f9ac7
CIFAR-10	55e0bcf8f4474c19

The split index arrays are reconstructed from the loader specification, not recovered as an independently archived execution-time split file. The MNIST and Fashion-MNIST tokens agree because their official pool sizes and index rules agree; their images are different. A compressed copy is supplied as `results/make50/reconstructed_split_indices.npz`. TwoNN uses Euclidean nearest-neighbour distances, excludes zero first distances and the empirical-CDF endpoint, and fits the log survival curve against the log distance ratio through the origin. The MLE cross-check averages local reciprocal mean-log-distance-ratio estimates over valid points at $k = 10$. Both estimates are floored at one in the local implementation.

Table A6. FONDUE short-training epochs and calibration time (once per dataset–context, separate from selector time). dSprites uses the source choice without calibration. A stable flag for the remaining datasets means consecutive calibration predictions agreed.

Dataset	β	Epochs	Calibration (s)	Stable
MNIST	0.25	2	9.64	yes
MNIST	0.5	2	9.55	yes
MNIST	1	1	4.14	yes
MNIST	2	2	8.12	yes
MNIST	4	2	9.38	yes
Fashion-MNIST	0.25	1	4.16	yes
Fashion-MNIST	0.5	1	3.64	yes
Fashion-MNIST	1	2	9.07	yes
Fashion-MNIST	2	3	13.70	yes
Fashion-MNIST	4	1	3.80	yes
dSprites	0.25	1	0.00	yes
dSprites	0.5	1	0.00	yes
dSprites	1	1	0.00	yes
dSprites	2	1	0.00	yes
dSprites	4	1	0.00	yes
CIFAR-10	0.25	3	30.34	yes
CIFAR-10	0.5	2	15.56	yes
CIFAR-10	1	2	22.72	yes
CIFAR-10	2	3	29.69	yes
CIFAR-10	4	4	42.98	yes

Appendix B.1. Pruning and Relevance Adaptations

The GECO adaptation uses Adam at 10^{-3} , batches of 128 in stored order, 300 epochs and hard-concrete parameters $\gamma = -0.1$, $\zeta = 1.1$ and temperature 2/3. Initial gate log-odds are 2. The constraint is summed squared reconstruction error minus $D_{\text{input}} T_{D,s}$, with moving-average coefficient 0.99. A squared softplus multiplier is clamped to $[10^{-6}, 10^6]$ and updated by ascent on that moving-average violation. KL is gate-masked; the expected gate penalty is enabled only when the moving-average constraint is satisfied. The final readout counts deterministic gates above 0.5. Its training constraint uses squared error even on dSprites; the transferred ordinary VAE there uses Bernoulli reconstruction. Consequently the candidate likelihood convention does not imply identical pruning-stage objectives.

Table A7. Raw retained counts versus transferred ordinary-VAE widths at the primary context. Constraint denotes the GECO training moving average at the final update; Q denotes final ordinary-VAE validation quality. They evaluate different models and quantities. ARD has no GECO constraint.

Dataset	Adaptation	Raw count	Final width	Constraint	Q
MNIST	GECO	7.2 ± 0.4	6.4 ± 0.9	0/5	5/5
MNIST	ARD	6.0 ± 0.0	6.0 ± 0.0	—	5/5
Fashion-MNIST	GECO	4.2 ± 0.4	4.0 ± 0.0	1/5	5/5
Fashion-MNIST	ARD	3.8 ± 0.4	3.6 ± 0.9	—	4/5
dSprites	GECO	64.0 ± 0.0	64.0 ± 0.0	0/5	5/5
dSprites	ARD	9.2 ± 1.5	8.4 ± 1.7	—	0/5
CIFAR-10	GECO	20.0 ± 1.0	19.2 ± 4.4	5/5	2/5
CIFAR-10	ARD	14.2 ± 0.4	16.0 ± 0.0	—	0/5

The ARD adaptation reserves the first 10% (1800 examples) of its training array for prior estimation and trains on the remainder. It updates $p_j = \text{mean}(\mu_j^2 + \sigma_j^2)$ every ten epochs and after training, clipping to $[10^{-6}, 10^6]$. This uses analytic posterior second moments in a fixed subset, rather than a periodically resampled encoded-data pool. Selection

training uses reconstruction plus Gaussian KL at unit weight. On the first 512 training examples it computes

$$w_j = \text{mean}_{x,k} [g_k(\mu(x) + \sqrt{p_j} e_j) - g_k(\mu(x))]^2, \quad q_j = w_j p_j, \quad (\text{A1})$$

and keeps the smallest number of descending scores reaching 99% of their total. The finite difference is not divided by its squared step. Locally it already contains a factor p_j , so multiplication by p_j again differs from a Jacobian-squared times variance score. The stored counts are therefore labelled an *ARD adaptation*; this difference cannot be removed by correcting the prose. The Gaussian approximation alone is not an original-source deviation. Identical relevance arrays and counts across contexts are checked in the numerical audit; they represent repeated fixed settings, not a β sweep.

The FONDUE description-based classifier was checked against the public repository https://github.com/bonhemi/VAE_learning_dynamics at commit f6d6e66b536bcbe635392d3438338949d4dc8730. The local implementation uses the stated posterior-variance categories with one exponentiation and TwoNN; equivalence to the public execution path and the original experiment configuration is not asserted. As a historical reference, the source reports dSprites selection 12.2 ± 0.4 at one epoch [4], versus 4.2 ± 0.4 here at $\beta = 1$. This unmatched-setting difference is not proof of an implementation error, nor a reproduction of the original benchmark.

Appendix B.2. Threshold-Only and Quality-Margin Analyses

The retrospective predictive analysis uses Python 3.11.11, NumPy 2.3.2 and scikit-learn 1.7.1. Table A8 documents probability saturation in this reconstruction. The original predictor's complete environment and row-level probabilities were not saved, so its exact floating-point results cannot be verified from the summary alone. Row-level decision scores and probabilities for the current reconstruction are saved in `results/make50/au_predictions.csv`.

Table A8. Numerical probability saturation in the current predictor reconstruction. Entries count predictions rounded to exactly one, before AUROC calculation. Decision scores preserve ordering among these rows.

Evaluation condition	Base	Base + AU	Base + random
dSprites	55/55	44/55	49/55
Fashion-MNIST	0/55	0/55	0/55
CIFAR-10	45/45	40/45	36/45

The main AU-threshold figure uses identical checkpoints for all three δ values. Across the full ladder, the fractions of CIFAR-10 grid/seed candidates with AU equal to capacity are given below. These counts use ten widths and five seeds per β ; their rows are not 50 independent datasets.

Table A9. CIFAR-10 (Gaussian): percentage of the 50 width/seed candidates with AU equal to capacity. Thresholds are re-read on the same checkpoints within each column.

δ	$\beta = .25$	.5	1	2	4
0.001	100	100	100	98	60
0.01	60	50	36	22	10
0.1	48	38	32	20	10

For a separate retrospective quality-margin check, the complete cached grid at $\beta = 1$ is held fixed and the smallest satisfying width is read out for each margin, without retraining

or changing the anchor. Table A10 reports this full-grid criterion, not the path or cost of any restricted-window selector. No FONDUE-threshold or ARD-99% sensitivity training was executed, so no response of those rules to their own readout thresholds is claimed.

Table A10. Smallest quality-satisfying cached width at $\beta = 1$ as relative margin changes; five seed means $\pm$ SD. The absolute floor 0.001 times validation variance remains fixed. Gaussian likelihood except Bernoulli dSprites.

Dataset	1%	5%	10%	25%	50%
MNIST	6.0 $\pm$ 0.0	6.0 $\pm$ 0.0	6.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0
Fashion-MNIST	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	4.0 $\pm$ 0.0	2.0 $\pm$ 0.0
dSprites	51.2 $\pm$ 17.5	36.0 $\pm$ 16.5	20.0 $\pm$ 4.0	12.0 $\pm$ 0.0	9.6 $\pm$ 0.9
CIFAR-10	41.6 $\pm$ 8.8	32.0 $\pm$ 0.0	24.0 $\pm$ 0.0	16.0 $\pm$ 0.0	16.0 $\pm$ 0.0

Appendix C. Synthetic Scale and Ablation Diagnostic

Swiss-roll points are $(t \cos t, h, t \sin t)$ with $t \sim U[1.5\pi, 4.5\pi]$ and $h \sim U[0, 1]$. Torus points are $((3 + \cos \phi) \cos \theta, (3 + \cos \phi) \sin \theta, \sin \phi)$ with independent angles uniform on $[0, 2\pi]$. Each has 5000 examples, generation seed zero, and independent Gaussian coordinate noise with SD 0.01. The first 4500 are used for fitting and the remaining 500 for the reported diagnostic. Per-coordinate standardisation in the stored experiment was fitted to all 5000 before splitting. This is disclosed as preprocessing leakage; the results are descriptive, not clean held-out test estimates.

The FC VAE uses widths 128 and 64, reversed in its decoder, ReLU hidden activations, a linear output, Adam at 10^{-3} , batch 128, 300 epochs, $\beta = 1$ and seeds 42, 123 and 777. At the last checkpoint, coordinates with posterior-mean variance at most 0.01 on the 500 diagnostic examples are zeroed and MSE is recomputed on those same examples. Table A11 reports the standardized runs. There is no separate test set in this stored experiment.

Table A11. Synthetic scale/ablation diagnostic, Gaussian likelihood, $\beta = 1$, $\delta = 0.01$, three training seeds. Data were standardized before the split; these are descriptive validation results, not leakage-free generalization estimates. Changes are mean per-run relative MSE changes (%).

Data	m	AU	MSE before	MSE after	Change
Swiss roll	1	1.0 $\pm$ 0.0	0.3511 $\pm$ 0.0267	0.3511 $\pm$ 0.0267	+0.000
Swiss roll	2	2.0 $\pm$ 0.0	0.1997 $\pm$ 0.0177	0.1997 $\pm$ 0.0177	+0.000
Swiss roll	3	3.0 $\pm$ 0.0	0.1407 $\pm$ 0.0018	0.1407 $\pm$ 0.0018	+0.000
Swiss roll	4	3.0 $\pm$ 0.0	0.1404 $\pm$ 0.0136	0.1406 $\pm$ 0.0135	+0.147
Swiss roll	6	3.0 $\pm$ 0.0	0.1365 $\pm$ 0.0023	0.1368 $\pm$ 0.0022	+0.183
Swiss roll	8	3.0 $\pm$ 0.0	0.1437 $\pm$ 0.0089	0.1440 $\pm$ 0.0089	+0.165
Torus	1	1.0 $\pm$ 0.0	0.3217 $\pm$ 0.0163	0.3217 $\pm$ 0.0163	+0.000
Torus	2	2.0 $\pm$ 0.0	0.1654 $\pm$ 0.0242	0.1654 $\pm$ 0.0242	+0.000
Torus	3	3.0 $\pm$ 0.0	0.1083 $\pm$ 0.0036	0.1083 $\pm$ 0.0036	+0.000
Torus	4	3.0 $\pm$ 0.0	0.1124 $\pm$ 0.0050	0.1124 $\pm$ 0.0050	+0.063
Torus	6	3.0 $\pm$ 0.0	0.1095 $\pm$ 0.0035	0.1096 $\pm$ 0.0035	+0.107
Torus	8	3.0 $\pm$ 0.0	0.1104 $\pm$ 0.0069	0.1105 $\pm$ 0.0069	+0.156

Both noise-free generating surfaces have intrinsic dimension two and are represented here in three ambient coordinates. A Swiss-roll patch does not thereby have minimum embedding dimension three. Observing three active coordinates under this loss is not a topological theorem. The earlier manuscript said its synthetic coordinates were standardized; the implementation audit found a mismatch between that description and the executed preprocessing. On the unstandardized Swiss roll the height coordinate has much smaller variance than the other coordinates, making its omission inexpensive in squared error. This scale observation does not alone prove the complete causal explanation of every earlier AU result.

Appendix D. Reproducibility Inventory and Remaining Records

The reaggregation entry point is `src/make50/rebuild_review.py`; it reads saved JSON and configuration files and writes numerical tables, six figures and the paired predictive analysis without VAE retraining. The manuscript assembly entry point is `src/make50/build_manuscript.py`. The generated main TeX embeds its table contents and references the figure PDFs. Compile from the project root with `pdflatex` repeatedly until references settle.

Table A12. Source-to-output map for the present revision. Paths below are relative to the project root.

Evidence	File or directory
Configuration	<code>config/experiment_config_MAKE49.json</code>
Ordinary method records	<code>results/make49/stage4_methods.json</code>
Pruning records	<code>results/make49/stage7_b7b8.json</code>
Candidate metrics and latent variances	<code>results/make49/stage4_cache/</code>
Train/validation and readout trajectories	<code>results/make49/stage3_e7a_e2.json</code>
ID representation diagnostics	<code>results/make49/stage5_e3_reference.json</code>
Original AU prediction result	<code>results/make49/stage5_e4_au_value.json</code>
Synthetic runs	<code>results/make49/stage6_e6_synthetic.json</code>
All audited method rows	<code>results/make50/method_results.csv</code>
Paired AU analysis	<code>results/make50/au_paired.json</code>
Snapshot identities	<code>results/make50/source_manifest.json</code>
Figures	<code>results/figures/MAKE50/</code>

Method-level row identifiers are (dataset, method, candidate seed, β or context). Candidate keys also contain width, requested epochs and stopping option. Every configuration/dataset–method–context cell in the main method archive has five seeds; this is checked during reaggregation. The ARD repetition does not create five distinct selection-weight conditions. Individual early off-grid cache entries are not all retained, although their final metrics and training traces remain in the method records. The original test predictions, full model checkpoints and independent ID/evaluation timings are not reconstructed from missing files.

The local change record documents the initial specification on 12 September; the switch to Bernoulli dSprites after collapse diagnostics; the move of ID estimation from an impossible 10,000-example validation sample to training data; trajectory runs without early stopping; and the 13 September reference-width, run-accounting and degenerate-output clarifications. The random-feature control was added after the primary predictive analysis. In this revision, anchor-cost completion, paired AU inference, quality-margin rereading and explicit implementation auditing are retrospective. None is relabelled as a publicly preregistered test.

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